

CHAPTER 13 Chromatography

The term 'chromatography' refers to a set of techniques that can be used to separate and analyse the components in a mixture. Separation is performed by passing the mixture through a medium that allows the components of the mixture to move through at different rates.

In this chapter you will learn about the basic principles of chromatography. You will also learn how these basic principles are applied in the instrumental techniques of thin-layer chromatography (TLC), high-performance liquid chromatography (HPLC) and gas chromatography (GC) which allow extremely sensitive and rapid analysis of the components in a mixture.

Science as a Human Endeavour

- Chromatographic techniques, including thin-layer chromatography (TLC), gas chromatography (GC), and high-performance liquid chromatography (HPLC), are used to determine the components of a wide range of mixtures in various settings. The decision to use a particular chromatographic technique depends on a number of factors, including the properties of the substances being separated, the amount of substance available for analysis and the sensitivity of the equipment. Chromatographic techniques have a wide range of analytical and forensic applications, including monitoring air and water pollutants, drug testing of urine and blood samples, and testing for food additives and quality.

Science understanding

- data from chromatography techniques, including thin-layer chromatography (TLC), gas chromatography (GC), and high-performance liquid chromatography (HPLC), can be used to determine the composition and purity of substances; the separation of the components is caused by the variation in strength of the interactions between atoms, molecules or ions in the mobile and stationary phases

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13.1 Principles of chromatography

Chromatography is a widely-used technique used to separate and analyse the substances present in a mixture. It can be used to analyse numerous inorganic and organic substances such as contaminants in water, toxic gases in air, additives and impurities in food, and drugs present in blood.

In this section, you will learn about the underlying principles of chromatography and also about three simple forms of the technique called **paper chromatography**, **thin-layer chromatography** and **column chromatography**.

CHEMISTRY IN ACTION

Drug testing in sport

Since competitive sports began, some people have sought to gain an advantage by whatever means available. For example, athletes in ancient Greece and Rome consumed specially prepared lizard meat in the hope that it would improve their performance and give them a competitive edge.

Today, drug problems in sport usually result from the inappropriate use of therapeutic substances and manipulation of substances such as hormones and electrolytes that occur naturally in an animal's or athlete's system. In horse racing, the use of illegal 'go-fast' or 'stopper' drugs is now relatively rare because of their ease of detection by an analytical technique called gas chromatography.

Gas chromatography is an extremely sensitive type of chromatography. Urine samples are routinely taken from athletes competing in major events to ensure that the athletes are not benefiting from the use of illegal, performance-enhancing drugs (Figure 13.1.1).

These samples are analysed by gas chromatography in combination with a mass spectrometer as the detector. You will learn more about gas chromatography later in this chapter.



FIGURE 13.1.1 A chemist prepares samples from athletes for drug testing



FIGURE 13.1.2 The pattern of bands produced when this stick of chalk is dipped in black ink and then placed in water is called a chromatogram.

HOW CHROMATOGRAPHY WORKS

All methods of chromatography have a:

- **stationary phase**
- **mobile** (moving) **phase**.

You can perform a simple chromatography experiment by dipping the end of a stick of chalk into water-soluble black ink and then standing the chalk in a beaker containing a small amount of water, as seen in Figure 13.1.2.

As the water carries the ink up the chalk, you will see that the ink separates into bands of different colours. Each band contains one of the substances, or **components**, present in the ink mixture. The pattern of bands or spots is called a **chromatogram**.

In this simple chalk-and-ink exercise, the stationary phase is the chalk and the mobile phase is the water.

As the components in the ink are swept upwards over the stationary phase by the solvent, they undergo a continual process of **adsorption** onto the solid stationary phase, followed by **desorption** and dissolving into the mobile phase. The ability of the components to stick to the stationary phase will depend upon the polarity of the stationary phase and the component molecules. Similarly, the attraction of the components to the solvent molecules is determined by their polarity.

The rate of movement of each component depends mainly upon:

- how strongly the component adsorbs onto the stationary phase
- how readily the component dissolves in the mobile phase.

The components separate because they undergo these two processes to different degrees. In the example involving the chalk and ink shown in Figure 13.1.2, the blue dye in the ink is more soluble in the mobile phase than the red dye, and bonds less strongly with the stationary phase than the red dye. Therefore the blue dye in the ink has moved faster up the piece of chalk than the red dye, resulting in their separation.

i The term 'adsorption' describes the adhesion of molecules or substances to the surface of a solid or liquid. The stationary phase in chromatography is either a solid or a liquid. The mobile phase is in either a liquid or a gas state.

Water, the mobile phase, is a polar solvent. The blue dye moves more quickly with the water up the stationary phase than the red dye, indicating that the blue dye contains compounds whose molecules are more polar than those in the red dye.

PAPER AND THIN-LAYER CHROMATOGRAPHY

In the laboratory, paper chromatography is performed on high-quality absorbent paper, similar to filter paper, as the stationary phase. Figure 13.1.5 shows paper chromatography in use for the separation of the components of different inks. Thin-layer chromatography (TLC) is very similar to paper chromatography. In this case the stationary phase is a thin layer of a fine powder, such as alumina (aluminium oxide), spread on a glass or plastic plate. Both techniques are useful for **qualitative analysis**, in which the chemicals that are present in a mixture are identified.



FIGURE 13.1.5 Paper chromatography can be used to separate mixtures of inks into their separate components.

CHEMFILE

Colour writing

While studying the coloured materials (pigments) in plants, Russian botanist Mikhail Tsvet (Figure 13.1.3) developed the separation technique known as chromatography. The word chromatography means 'colour writing' and comes from the Greek *khromatos*, meaning 'of colour', and *graphos*, meaning 'written'.



FIGURE 13.1.3 Mikhail Tsvet was the inventor of chromatography.

Passionate about botany, Tsvet discovered that different pigments appeared as different coloured bands. In 1903, chlorophyll and xanthophyll were the only known plant pigments. Tsvet produced chromatograms similar to the one shown in Figure 13.1.4 and discovered two forms of chlorophyll and eight other pigments.

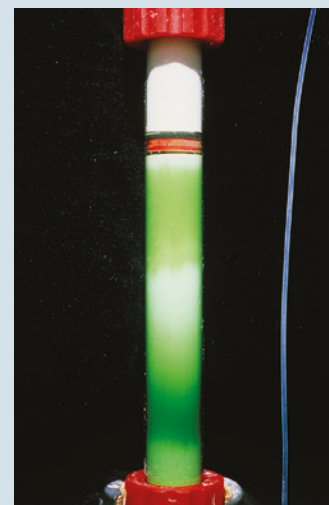


FIGURE 13.1.4 Chromatogram of a plant sample similar to the chromatograms obtained by Mikhail Tsvet

Chromatography is such an important technique that two Nobel prizes have been awarded for research based mainly on this analytical method.

Paper and thin-layer chromatography in practice

In both paper and thin-layer chromatography, a small spot of the solution of the sample to be analysed is placed on one end of the chromatography paper or plate.

The position of this spot is called the **origin**. The paper or plate is then placed in a container with solvent. The origin must be a little above the level of the solvent so that the components can be transported up the paper or plate and not dissolve into the liquid in the container. As the solvent rises up the paper or plate, the components of each sample separate depending on their attraction to the stationary phase and their solubility in the solvent, as shown in Figure 13.1.6.

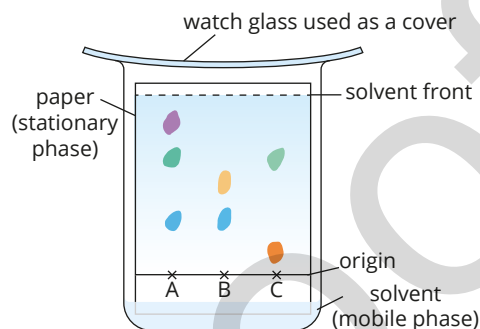


FIGURE 13.1.6 Paper chromatography of three different food colours (A, B and C)

Identifying the components of a mixture

The components in a mixture can be identified by chromatography in one of two ways:

- 1 by including standards of known chemicals on the same chromatogram as the unknown sample and comparing the resulting positions of the unknown components with those of the known samples.
- 2 by calculating the **retardation factor** (R_f) of the sample and comparing these with the R_f values of known samples.

Method 1: Using standards

In this method, you need to know what chemicals might be present in the sample. For example, if you wish to find out whether a vitamin tablet contains vitamins A and D, a sample of the vitamin tablet can be placed alongside samples of each of vitamins A and D on the same chromatogram. If spots from the tablet sample move the same distance from the origin as the spots from the pure samples, then the tablet is likely to contain the vitamins.

The sample and standards are 'run' on the same chromatogram because the distances moved from the origin will depend on the distance moved by the solvent front. The further the solvent front is allowed to travel, the further the spots travel.

Method 2: Calculating R_f values

Another way of identifying the components of a mixture is by comparing the distance they travel up the stationary phase to the distance travelled by the solvent front. This is expressed as a retardation factor, R_f , for a component:

$$R_f = \frac{\text{distance the component travelled from the origin}}{\text{distance the solvent front travelled from the origin}}$$

You can see from the chromatogram in Figure 13.1.7 that:

- R_f values will always be less than one
- the component most strongly adsorbed onto the stationary phase moves the shortest distance and has the lowest R_f value.

Each component has a characteristic R_f value for the conditions under which the chromatogram was obtained. By comparing the R_f values of components of a particular mixture with the R_f values of known substances determined under identical conditions, you can identify the components present in a mixture.

i 'Solvent front' is the term used to describe the movement of the solvent during chromatography. It is visible as the wet moving edge of the solvent as it travels along the stationary phase.

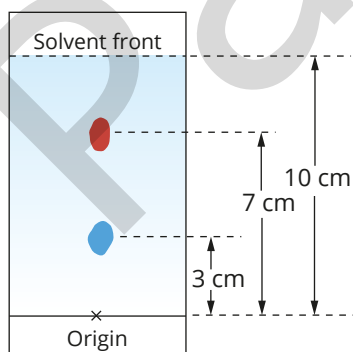


FIGURE 13.1.7 A chromatogram of a sample that consists of two components. The distances from the origin enable R_f calculations.

In this method, the actual distance moved by the solvent front is no longer critical as the proportion of the distance moved from the origin (the R_f value) stays the same provided the conditions under which the chromatogram is obtained are the same.

This means the R_f values of unknown spots can be compared against a table of R_f values of common materials. However, changes in the temperature, the type of stationary phase, the amount of solvent vapour around the paper or plate and the type of solvent will all change the R_f value for a particular chemical.

i For the R_f value of a component to be comparable to a set of standard R_f values, they must be determined under identical conditions.

Worked example 13.1.1

CALCULATING R_f VALUES

Calculate the R_f value of the red component in Figure 13.1.7.	
Thinking	Working
Record the distance the component has moved from the origin and the distance the solvent front has moved from the origin.	Distance from origin of red component = 7 cm Distance from origin of solvent front = 10 cm
$R_f = \frac{\text{distance of component from origin}}{\text{distance of solvent front from origin}}$	$R_f(\text{red component}) = \frac{7}{10}$ $= 0.7$

Worked example: Try yourself 13.1.1

CALCULATING R_f VALUES

Calculate the R_f value of the blue component in Figure 13.1.7.

In chromatograms of inks, plant pigments and food dyes, the components can be seen easily; however, most compounds are colourless and must be made visible. Many organic compounds fluoresce and appear blue when viewed under ultraviolet light. For other compounds, the chromatogram can be sprayed with a chemical that reacts to form coloured or fluorescent compounds. For example, a chemical called ninhydrin reacts with amino acids to give blue and brown coloured compounds that are easily detected on the chromatograph.

The choice between paper and thin-layer chromatography depends upon the sample being analysed. Table 13.1.1 lists the advantages of each method.

TABLE 13.1.1 A comparison of the advantages of paper and thin-layer chromatography

Paper chromatography	Thin-layer chromatography
cheap	detects smaller amounts
little preparation	better separation of less polar compounds
more efficient for polar and water-soluble compounds	corrosive materials can be used
easy to handle and store	a wide range of stationary phases is available

For a particular combination of stationary phase and mobile phase, many different chemicals may have similar R_f values. Paper and thin-layer chromatography only provide a guide to the identity of a chemical. Further testing using high-performance liquid chromatography (HPLC) or gas chromatography (GC) may be required to confirm the identity of a chemical. Section 13.2 will introduce you to HPLC and GC, and how they can be used to determine the presence *and* concentration of chemicals in a mixture.

Thin-layer chromatography in forensic analysis

Many forensic investigations use thin-layer chromatography (TLC) to analyse materials recovered from the scene of an investigation. TLC is useful in detecting chemicals of forensic concern, including chemical weapons, explosives and illicit drugs.

Scientists can use a computerised and portable TLC machine that can be taken anywhere and used to analyse multiple samples at a time. These analyses can be completed within 30 minutes, allowing analyses to be performed at the scene rather than in a dedicated laboratory.

The technique is inexpensive, reliable, fast, and easy to perform. The method was refined in the late 1960s so that it could reliably measure the amounts of compounds.

High-performance thin-layer chromatography

High-performance TLC (HPTLC) is an enhanced form of TLC that automates the different steps, increases the resolution achieved and allows more accurate measurements of the quantities of components present in a sample. It provides better resolution of chemically similar compounds than conventional TLC, and requires smaller samples (Figure 13.1.8). However, HPTLC requires specialised analysis equipment, and so is still not as popular or widespread as conventional TLC.

When a sample undergoes analysis, there may be uncertainty in the size of the drop of sample and its position at the origin when deposited by hand. This uncertainty has been overcome in HPTLC by replacing the manual method with piezoelectric devices and inkjet printers.

The plate can also be developed using two different solvents. Initially, the sample is applied to the plate and developed with the first solvent. Then the plate is rotated by 90° and developed with a second solvent. This procedure increases the resolution of the separation and produces more accurate R_f values.

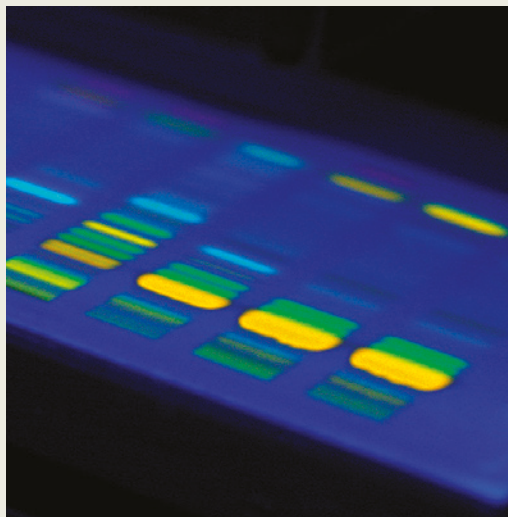


FIGURE 13.1.8 These glowing bands, seen under fluorescent lighting, are due to the components of mixtures separated on a high-performance thin-layer chromatography plate.

COLUMN CHROMATOGRAPHY

Figure 13.1.9 shows another form of chromatography—column chromatography—which can also be used to separate the components of a mixture.

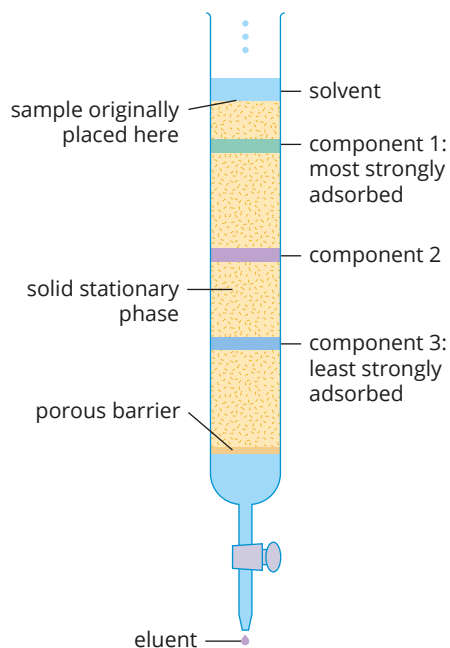


FIGURE 13.1.9 Column chromatography is used to separate the components in a mixture.

The solid stationary phase is packed into a glass column. The sample mixture is applied carefully to the top of the packed solid, and a solvent, which acts as the mobile phase, is dripped slowly onto the column from a reservoir above. A tap at the bottom of the column allows the solvent, which is called the **eluent**, to leave the column at the same rate as it enters it at the other end. Figure 13.1.10 shows how column chromatography can be used to separate and collect different components in a black ink.

Add black ink.
As solvent is added,
separation occurs
and components
are collected.

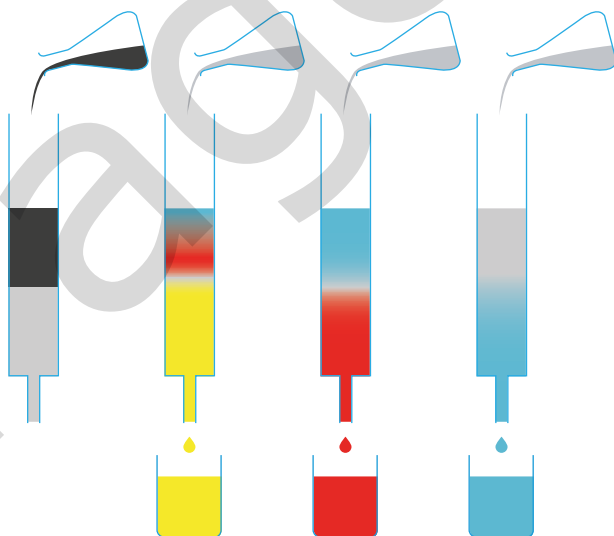


FIGURE 13.1.10 Column chromatography can be used to separate and collect the components in black ink. The yellow dye in the ink is more soluble in the mobile phase and adsorbs less strongly to the stationary phase than the red and blue dyes. The yellow dye in the ink moves faster through the column than the red and blue dyes, resulting in separation of the dyes.

13.1 Review

SUMMARY

- Chromatography is a technique commonly used to separate and identify the components in a mixture.
- Paper and thin-layer chromatography are simple forms of this technique.
- All chromatographic techniques involve a mobile phase and a stationary phase.
- Components separate during chromatography as a consequence of how strongly they adsorb to the stationary phase and desorb back into the mobile phase.
- Components in a mixture have differing affinities for the mobile and stationary phases.
- In paper and thin-layer chromatography, the components in a mixture are identified by comparison with known standards and determination of R_f values.

KEY QUESTIONS

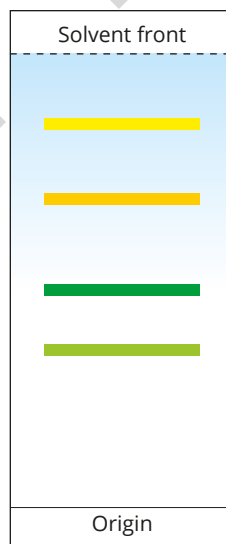
- 1 Paper chromatography is used to separate the pigments in a plant leaf. A spot of each pigment is placed on a sheet of chromatography paper and the chromatogram is run inside a closed jar that is partly filled with ethanol. Which one of the following options is the mobile phase in this instance?

- A** pigments
B ethanol
C paper
D ethanol vapour in the closed jar

- 2 Match each term with its correct description.

Term	Description
adsorption	the breaking of the attraction between a substance and the surface to which the substance is adsorbed
desorption	the components of a mixture undergo adsorption to this phase
components	the different compounds in the mixture, which can be separated by chromatography
polar molecule	the attraction of one substance to the surface of another
mobile phase	a molecule that acts as a dipole; it has one or more polar covalent bonds, with the charge being distributed asymmetrically
stationary phase	the solvent that moves over the stationary phase in chromatography

- 3 An extract from a plant was analysed by thin-layer chromatography with a non-polar solvent. The chromatogram obtained is shown in the figure below. The table below gives the R_f values of some chemicals commonly found in plants measured under the same conditions.



Chemical	R_f
xanthophyll	0.67
β -carotene	0.82
chlorophyll a	0.48
chlorophyll b	0.35
leutin	0.39
neoxanthin	0.27

- a** Using the figure above, measure and record the distance from the origin to the centre of each band, and the distance of the solvent front from the origin.
- b** Calculate the R_f value of each band.
- c** Compare R_f values for the bands with the R_f values in the table above and name the chemicals present in the extract.
- d** If water had been used as the solvent, would the chromatogram be likely to have a similar appearance? Explain your reasoning.

4 Phenacetin was once an ingredient in analgesic drugs, but it is not used now because it causes liver damage. It is soluble in chloroform. A chemist wishes to analyse a brand of analgesic by thin-layer chromatography to determine whether it contains phenacetin. Outline the steps in the analysis. (Assume that a sample of pure phenacetin is available to the chemist.)

5 Use the following terms to complete the summary of column chromatography. Not all terms are used.

adsorb

desorb

longest

mobile

shortest

stationary

During column chromatography, the components of the sample _____ onto the stationary phase and _____ into the liquid mobile phase. A component that adsorbs most strongly to the

_____ phase and is least soluble in the

_____ phase would be expected to take the

_____ time to pass through the column.

13.2 Advanced applications of chromatography

In Section 13.1, you learnt the principles of chromatography and how the techniques of paper, thin-layer and column chromatography can be used to separate the components in a mixture.

In these techniques, a liquid mobile phase is used to carry the components of the mixture past a solid stationary phase. The components undergo a continual process of adsorption to the stationary phase and desorption back into the mobile phase. The components of a mixture undergo the processes of adsorption and desorption to different degrees, depending on the strength of their attractions to the stationary phase and the mobile phase. As a result, the components separate as they move past the stationary phase.

In this section, you will learn how these principles are used as the basis for two modern analytical techniques, high-performance liquid chromatography and gas chromatography, which are highly sensitive and capable of detecting miniscule amounts of a compound.

HIGH-PERFORMANCE LIQUID CHROMATOGRAPHY

High-performance liquid chromatography (HPLC) is a chromatographic technique based on column chromatography that allows scientists to perform extremely sensitive analyses of a wide range of mixtures. It is commonly used for the separation and identification of complex mixtures of similar compounds, such as contaminants that are soluble in water, drugs in blood and hydrocarbons in oil samples.

i High-performance liquid chromatography (HPLC) is sometimes referred to as high-pressure liquid chromatography.

HPLC is now used routinely for environmental, pharmaceutical and industrial analyses. For example, it can be used to analyse for the presence and concentration of dioxins, insecticides, pesticides and oil spills in water. It is also used to determine the presence of pesticides in food or to detect the presence of drugs in blood.

Figure 13.2.1 shows the apparatus used for HPLC. The basic principles of HPLC are the same as for column chromatography described earlier. There are two main differences between simple column chromatography and HPLC.

- The particles in the solid used in the HPLC column are often 10–20 times smaller than those used in column chromatography, allowing more frequent adsorption and desorption of the components, giving much better separation of similar compounds.
- The small particle size used in HPLC creates a considerable resistance to the flow of the mobile phase and so the solvent is pumped through the column under high pressure.

A range of solids is available for use in HPLC columns, some with chemicals specially bonded to their surfaces to improve the separation of particular classes of compounds.

In HPLC, the components are usually detected by passing the eluent stream through a beam of ultraviolet (UV) light. Many organic compounds absorb UV light, so when an organic compound passes in front of the beam of light a reduced signal is picked up by a detector. The amount of light received by the detector is recorded on a chart that moves slowly at a constant speed or sent to a computer. The resulting trace is called a chromatogram. Each component forms one peak in the chromatogram.

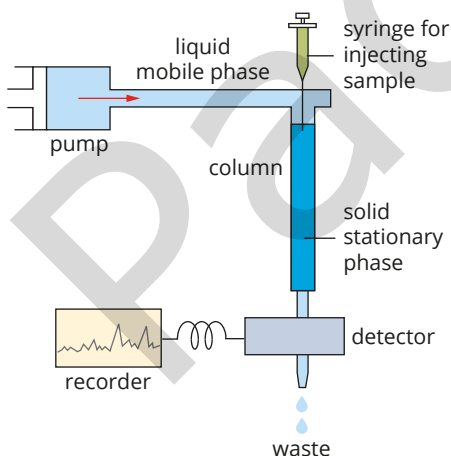


FIGURE 13.2.1 The construction of a high-performance liquid chromatograph

Figure 13.2.2 indicates how the chromatogram is produced as the components pass through the instrument. Figure 13.2.3 shows the HPLC instrument in use.

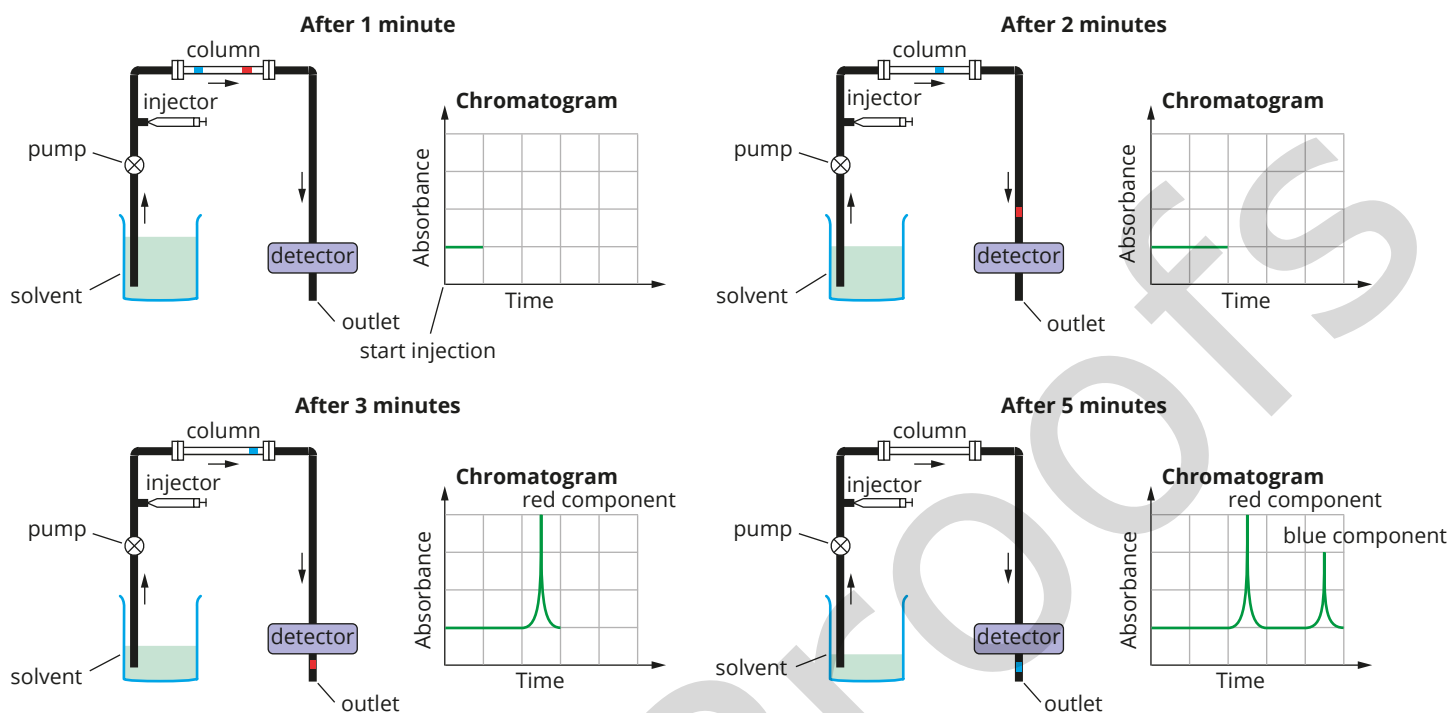


FIGURE 13.2.2 Separation of components of a sample using a high-performance liquid chromatograph



FIGURE 13.2.3 A research chemist using high-performance liquid chromatography to analyse the components in a mixture for medical research

GAS CHROMATOGRAPHY

The most sensitive of the chromatographic techniques is **gas chromatography (GC)**. It is capable of detecting as little as 10^{-12} g of a component. However, it is limited to components that can be readily vapourised without decomposing. Such compounds usually have relative molecular masses of less than 300. HPLC, on the other hand, can separate compounds with relative molecular masses that are as high as 1000 or even more.

i Gas chromatography is an extremely sensitive analytical technique. However, its use is limited to compounds that vaporise without decomposing (usually with relative molecular masses of less than 300).

The extreme sensitivity of GC makes it ideal for the analysis of trace contaminants in samples or for the detection of tiny amounts of very potent components. For example, the urine samples routinely taken from athletes competing in major events are analysed by GC to ensure that the athletes are not benefiting from the use of illegal, performance-enhancing drugs.

A generalised diagram of the equipment used in GC is shown in Figure 13.2.4.

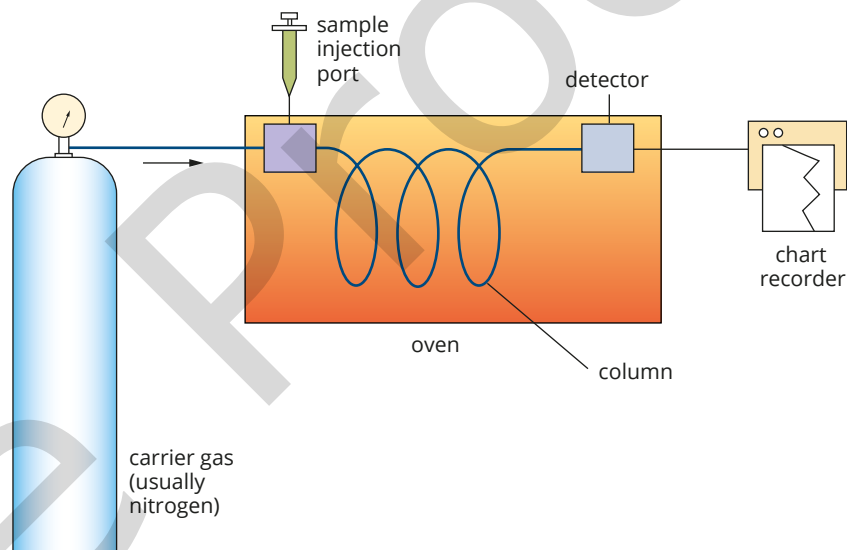


FIGURE 13.2.4 A gas chromatograph

CHEMFILE

The flame ionisation detector—an Australian invention

When the solid particles used in the column of a gas chromatograph are coated with a liquid hydrocarbon or ester with a high boiling point, the instrument is known as a gas–liquid chromatograph (GLC) and the liquid acts as the stationary phase.

Figure 13.2.5 shows one of the most useful detectors used in gas–liquid chromatography, the flame ionization detector, invented in 1957 by an Australian, Ian McWilliam.

In this detector, organic compounds leaving the column are burnt in a hydrogen–oxygen flame. Ions produced in the flame are attracted to electrodes and cause a current to flow. This current is used to indicate the presence of the component and, after calibration, the component's concentration in the original sample.



FIGURE 13.2.5 A flame ionisation detector

Gas chromatography has the following features:

- The mobile phase is an unreactive gas, often nitrogen, called the **carrier gas**.
- A small amount of sample is injected into the top of the column through an **injection port**. The injection port is heated to vaporise the sample, which is then swept into the column by the carrier gas.
- The column is a series of loops of glass or metal that has an internal diameter of about 4 mm and is about 2–3 m long in total. It is heated. The column can be packed with a porous solid coated with an ester or liquid hydrocarbon with a high boiling point, or it can be packed with an adsorbent solid such as silica gel or alumina. The solid acts as a stationary phase.

As the sample passes through the instrument, the components of the sample repeatedly interact with the stationary phase and are swept forward by the carrier gas. Components that adsorb least strongly to the stationary phase are swept out first by the gas. As the components emerge from the end of the column, they are sensed by the detector.

APPLICATIONS OF HPLC AND GC

Chemists use HPLC and GC in chemical analysis to answer two questions.

- What chemical is present in the sample? (qualitative analysis)
- How much of each chemical is present? (**quantitative analysis**)

i HPLC and GC provide qualitative and quantitative analysis of components in a mixture. The size of the peaks is due to the amount of light absorbed by each component, and this can be used to calculate the relative amount of each component within the mixture.

Qualitative analysis

Qualitative analysis can be used to identify the components in a sample and determine its purity. A solution of a pure compound that is thought to be one of the components is injected into the HPLC or GC under the same conditions. The chromatogram of the standard is then compared with the chromatogram of the sample.

The time taken for a component to pass through the HPLC or GC column is called its **retention time (R_t)**, and is characteristic of the component for the conditions of the experiment. It is used in the same way as the retardation factor (R_f) in paper and thin-layer chromatography.

The same compound will have the same retention time if the conditions (temperature, mobile phase, stationary phase, flow rate, pressure etc.) remain the same. Retention times can therefore be used to identify the components causing the peaks on a gas or liquid chromatogram.

Example: Detecting pesticides

Pesticides can be used to prevent mould in food such as the lettuce used in salads. HPLC can be used to test waste water samples from a farm for the presence of two pesticides: vinclozolin and carbendazim.

The high-performance liquid chromatogram of a water sample is shown in Figure 13.2.6. The two component pesticide peaks, marked 1 and 2, are identified by obtaining chromatograms of known pure compounds under exactly the same conditions as the sample.

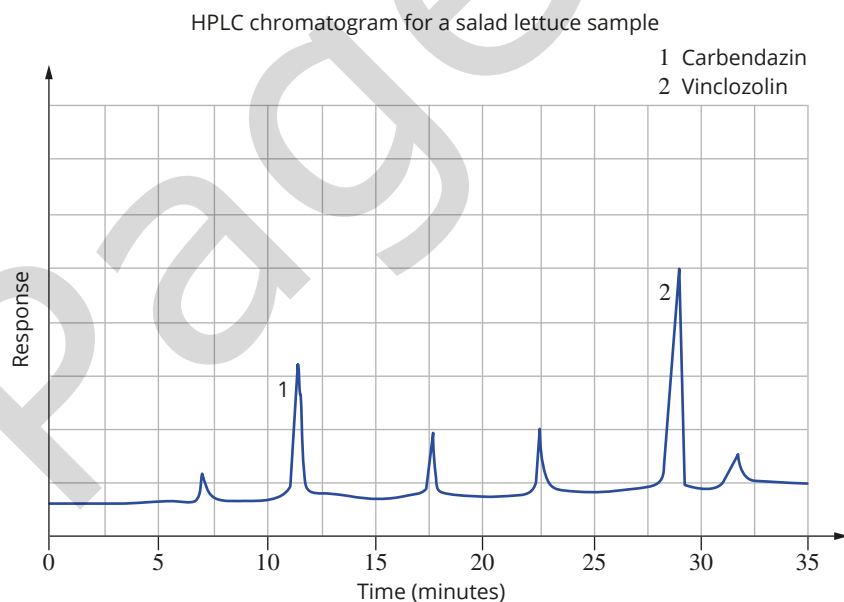


FIGURE 13.2.6 High-performance liquid chromatogram of a water sample from a farm

A compound can also be identified by adding a known compound to the sample (a procedure called ‘spiking’). The chromatogram in Figure 13.2.7 was produced when the water sample was spiked with carbendazin. Note that peak 1 is much bigger in the spiked sample. There are no extra peaks, indicating that carbendazin must have been component 1 in the sample.

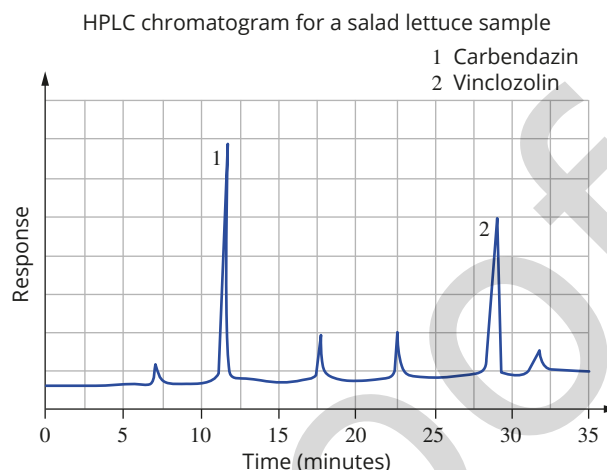


FIGURE 13.2.7 High-performance liquid chromatogram for a spiked water sample

Quantitative analysis

To determine the concentration of an individual component in a mixture, its peak area is compared with the peak areas of samples of the same chemical at known concentrations. A solution with an accurately known concentration is called a **standard solution**.

By plotting the peak areas against the concentrations of the standard solutions, a calibration curve can be drawn and used to determine unknown concentrations.

Worked example 13.2.1 shows how HPLC or GC can be used to find the concentration of a component in a mixture.

Worked example 13.2.1

CONCENTRATION OF A PESTICIDE

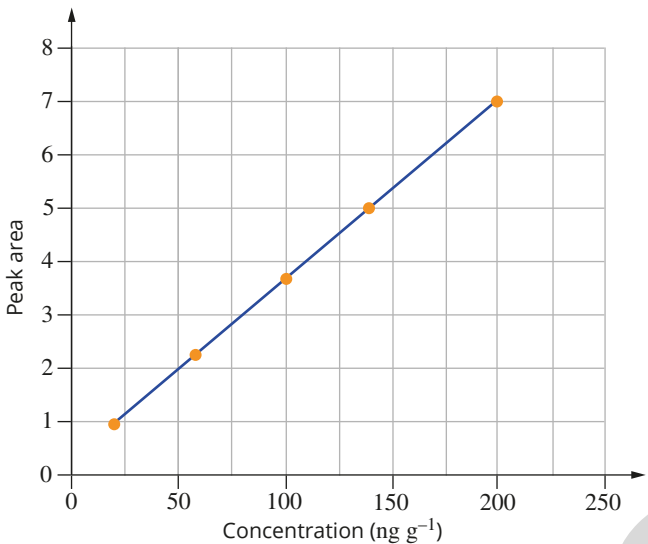
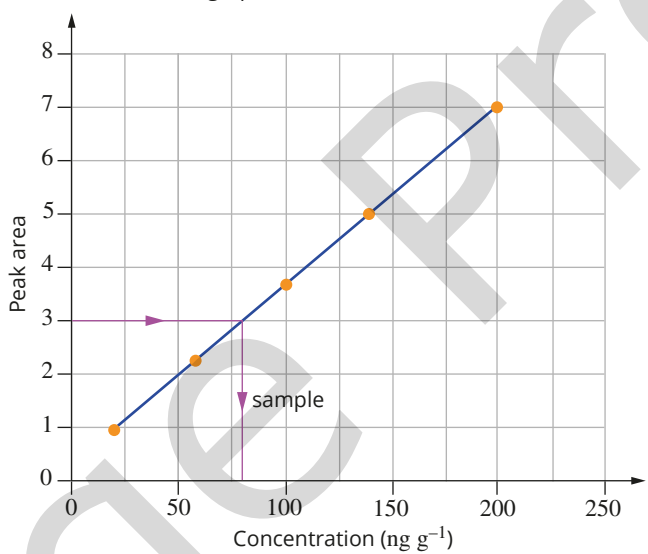
The concentration of the pesticide carbendazin in a sample of waste water from a farm was determined by HPLC. The chromatograms of a series of standards with accurately known concentrations of carbendazin were obtained under the same conditions as the sample.

The results from the chromatograms of the sample and the standards are shown in the following table. Determine the concentration of carbendazin in the sample of waste water. (Note that units of ng g^{-1} are used because the concentrations are so low— 200 ng g^{-1} is equivalent to just 0.2 ppm (parts per million) or 0.0000002% by mass.)

	Peak area	Concentration (ng g^{-1}); $1 \text{ ng} = 10^{-9} \text{ g}$
Standard 1	0.95	20
Standard 2	2.20	60
Standard 3	3.60	100
Standard 4	5.00	140
Standard 5	7.00	200
Sample	3.00	?

i The information that is obtained from a high-performance liquid chromatogram or a gas chromatogram is:

- Retention time, R_t – which can be used to identify components in a mixture.
- Peak area – which allows the concentration of an individual component in a mixture to be determined by comparing its peak area with the peak areas of samples with known concentrations of the same chemical.

Thinking	Working
Construct a calibration graph by plotting peak area versus concentration.	
Mark where the peak area of the sample lies on the calibration curve by tracing a horizontal line to the curve and then a vertical line down to the x-axis.	Calibration graph for carbendazin in salad lettuce 
Determine the concentration of the sample by reading the concentration value for the unknown solution from the x-axis.	The concentration of carbendazin in the water sample is 80 ng g^{-1} .

Worked example: Try yourself 13.2.1

CONCENTRATION OF A PESTICIDE

Procymidone is a pesticide that is used to prevent disease in oranges. Tests have been performed by HPLC to determine the concentration of this pesticide in a sample of orange juice.

The chromatograms of a series of standards with accurately known concentrations of procymidone were obtained under the same conditions as the sample.

The results from the chromatograms of the sample and the standards are shown in the following table.

Determine the concentration of procymidone in the sample of orange juice.

	Peak area	Concentration (mg kg ⁻¹)
Standard 1	10	0.5
Standard 2	20	1.0
Standard 3	30	1.5
Standard 4	40	2.0
Standard 5	50	2.5
Sample	15	?

CHEMISTRY IN ACTION

What's in your olive oil?

Gourmet cooks value olive oil (Figure 13.2.8) for its distinctive taste and aroma. Nutritionists favour it because it is rich in monounsaturated triglycerides, which are believed to lower blood cholesterol levels and reduce the risk of heart disease. But there is continuing frustration among Australian olive oil producers at the lack of regulation of olive oil sold in Australia. Imported olive oil has shown evidence of rancidity and, in some cases, substitution of less expensive oils such as corn, peanut and soybean for olive oil.



FIGURE 13.2.8 Pure olive oil is highly regarded for its aroma and taste, and because it is thought to reduce blood cholesterol levels.

The HPLC technique can be used by food scientists as the basis for an analytical test based on olive oil's unique composition. Chromatography separates the oils into their component triglycerides. An example of the chromatograms obtained for pure olive oil and an impure oil can be seen in Figure 13.2.9.

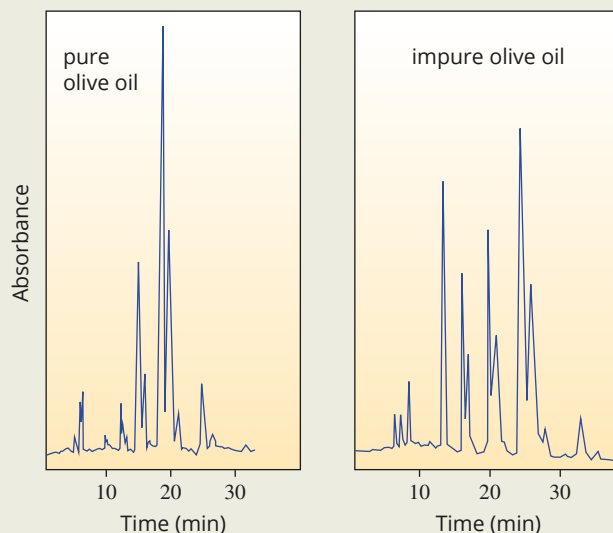


FIGURE 13.2.9 Peaks in the chromatogram of an impure oil indicate the presence of compounds found in corn oil, rather than olive oil.

COMPARING CHROMATOGRAPHIC TECHNIQUES

In Sections 13.1 and 13.2 you learned about different chromatographic techniques that can be used for analysis: paper and thin-layer chromatography, HPLC and GC. What factors do chemists consider when choosing which technique to use in a particular situation?

The decision to use a particular technique depends on a number of factors, including the:

- properties of the components being separated
- amount of sample available for analysis
- concentration of the component in the sample
- sensitivity of the technique
- time taken for analysis
- cost of equipment.

Some of the important features of the different chromatographic techniques are listed in Table 13.2.1. Chromatography is widely used for the drug testing of blood and urine samples, monitoring water and air pollution, and testing food quality.

TABLE 13.2.1 Features of different chromatographic techniques

Technique	Typical substances tested	Typical samples	Advantages	Disadvantages	Comments
paper and thin-layer chromatography	polar, water-soluble substances	drug detection; dyes in foodstuffs	very cheap, only basic laboratory equipment needed; easy to perform	poor precision and accuracy	samples need to be coloured or visible under UV light; otherwise, samples can be made visible using stains.
high-performance liquid chromatography	medium to high molecular mass organic compounds, e.g. pesticides, enzymes	foods, drugs, biological samples	high sensitivity and precision; small sample size; readily automated	moderately expensive instrument; trained technician needed to operate	samples must be soluble in a suitable solvent
gas chromatography	low molecular mass organic compounds, e.g. acetone, aspirin	water, gases, foods, drugs, biological samples	high sensitivity and precision; small sample size; readily automated	moderately expensive instrument; trained technician needed to operate	samples must be vaporised without breaking down

EXTENSION

Combining analytical techniques

Both GC and HPLC can be combined with another analytical technique, mass spectrometry (MS), which you learnt about in Chapter 2. Together, the techniques allow chemists to determine very small quantities and identify a wider range of materials, as well as gain valuable information about the structures of the compounds.

The advantage of the techniques of gas chromatography-mass spectrometry (GC-MS) and high-performance liquid chromatography-mass spectrometry (HPLC-MS) is that, by first passing the sample through a chromatograph, a complex sample can be separated into its many components. Each component then passes through the mass spectrometer, which can compare unknown mass spectra with an online library containing hundreds of thousands of different spectra.

GC-MS and HPLC-MS (Figure 13.2.10) have become essential techniques for many types of analyses.



FIGURE 13.2.10 An HPLC-MS instrument in a laboratory

13.2 Review

SUMMARY

- High-performance liquid chromatography (HPLC) and gas chromatography (GC) are very sensitive techniques used for qualitative and quantitative analysis.
- In HPLC, the mobile phase is liquid under pressure; in GC the mobile phase is a gas.
- In HPLC and GC, retention time is used to identify components in a mixture.
- The concentration of an individual component in a mixture can be determined by comparing its peak area on a chromatogram with the peak areas of samples with known concentrations of the same chemical analysed under identical conditions as the mixture.

KEY QUESTIONS

- 1 The retention time can be used in GC to determine:
A the identity of the chemical
B the concentration of a chemical
C the amount of a chemical in a sample
D all of the above.
- 2 The concentration of a substance is most accurately determined in HPLC by measuring:
A peak area
B peak height
C retention time
D R_f value.
- 3 Compounds A and B are equally soluble in water, but A is slightly more strongly adsorbed on silica powder. Compound C has a lower solubility in water than A and B. A sample containing A, B and C is mixed with water and injected into an HPLC instrument, which uses water as the solvent and silica as the stationary phase. List the order of the components as they are likely to emerge from the column.
- 4 Nitroglycerine patches are used to treat forms of heart disease. A sample from a patch was injected onto a gas-liquid chromatograph column. The peak corresponding to nitroglycerine had an area of 4.4 mm^2 . The peak areas for three standard solutions were also measured, as shown in the following table.
 - a Construct a calibration curve of peak area versus concentration of nitroglycerine.
 - b Determine the concentration of nitroglycerine in the sample in mg mL^{-1} .

Nitroglycerine standards ($\mu\text{g mL}^{-1}$)	Peak area (mm^2)
2	3.0
4	5.8
6	8.8

- 5 The ethanol content of waste water from a brewery can be analysed by HPLC. The peak areas produced from HPLC analysis of a sample of water and a number of standard solutions of ethanol are shown in the table below.

	Ethanol (%)	Relative peak area
Water sample	?	5.6
Standard 1	0.2	1.6
Standard 2	0.4	3.2
Standard 3	0.6	4.8
Standard 4	0.8	6.4

- a Construct a calibration curve of relative peak area against concentration of ethanol (%).
- b Determine the percentage of ethanol in the water sample, correct to 1 decimal place.

Chapter review

KEY TERMS

adsorption
carrier gas
chromatogram
chromatography
column chromatography
components
desorption

eluent
gas chromatography (GC)
high-performance (or
high-pressure) liquid
chromatography (HPLC)
injection port
mobile phase

origin
paper chromatography
qualitative analysis
quantitative analysis
retardation factor (R_f)
retention time (R_t)
standard solution

stationary phase
thin-layer chromatography

Principles of chromatography

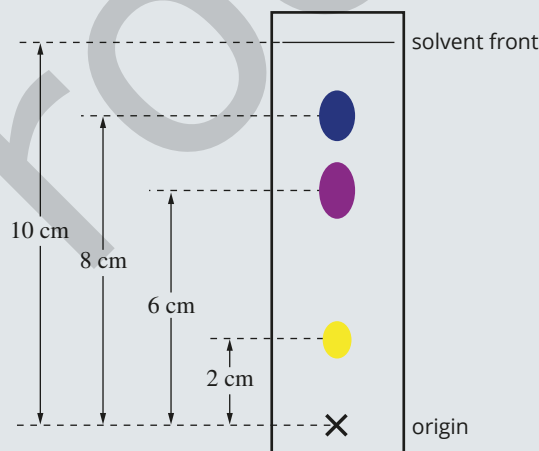
- 1 Use the following terms to complete the sentences about paper chromatography. Words can be used more than once.

above	mobile	phases
below	origin	sample
components	phase	stationary

In paper chromatography, the paper acts as the _____ phase. A small spot of a solution is placed at one end of the paper, called the _____. The sample solution contains a number of different coloured compounds, the _____. The paper is suspended so that the end with the spot is _____ the surface of the solvent. The solvent or _____ phase moves up the _____ phase. Different coloured spots are observed at various places on the paper, due to the separation of different _____.

- 2 a Use the terms 'adsorbed' and 'absorbed' correctly in each of the sentences below.
- Water was _____ by the towel as the wet swimmer dried himself.
 - A thin layer of grease _____ onto the cup when it was washed in the dirty water.
- b Explain the difference between the terms 'adsorbed' and 'absorbed'.
- 3 Describe how the rate of adsorption and desorption of a component that appears at the top of a paper chromatogram compares to that of a component that appears at the bottom of the chromatogram.
- 4 When a solution containing components X and Y is placed in a chromatography column, the rates at which X and Y move down the column are determined by several factors. Which one of the following factors does *not* directly affect the rate at which X and Y move down the column?
- the polarity of the molecules of X and Y
 - the attractions of X and Y to the stationary phase
 - the solubility of X and Y in the solvent
 - the length of the chromatography column

- 5 The chromatogram of a dye is shown below. Calculate the R_f values for each of the blue, purple and yellow components. Give answers to 1 decimal place.

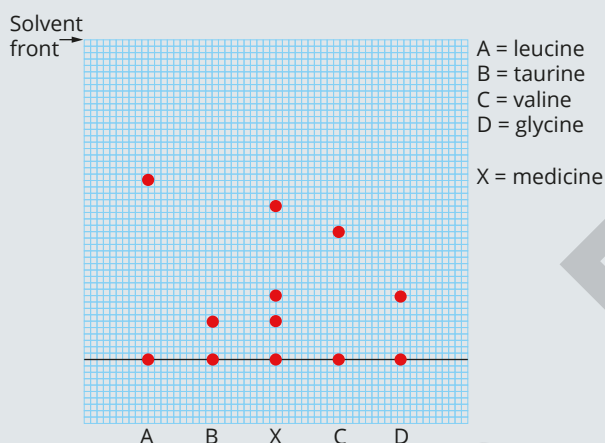


- 6 A sample of brown dye from a lolly is placed at the origin on a chromatography plate. The solvent front moves 9.0 cm from the origin. A blue component of the dye moves 7.5 cm and a red component 5.2 cm in the same time. Calculate the R_f values of the two components.
- 7 Consider the paper chromatogram of three food colours in Figure 13.1.6 on page xxx.
- Why must the level of the solvent be lower than the origin where spots of the mixture are originally placed?
 - Why are R_f values always less than one?
 - How many different components have been used to make colour A?
 - Which components present in colours B and C are also in colour A? Explain.
 - Which component of colour A is most strongly adsorbed on the stationary phase?
 - Calculate the R_f values of each component of colour C.

- 8 Thin-layer chromatography showed that the black dye used in a brand of writing ink contained blue, red, orange and yellow components. The R_f values of these substances in ethanol as solvent are 0.59, 0.32, 0.80 and 0.19, respectively.

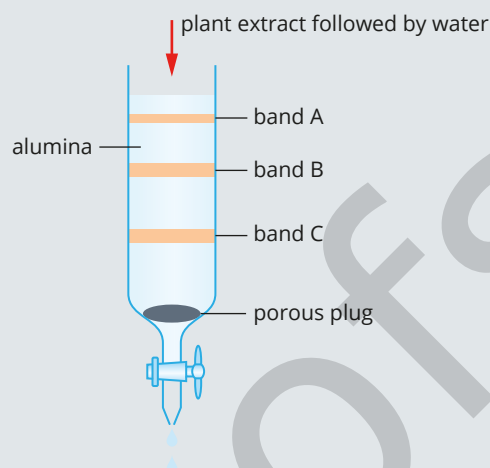
- How far apart would the blue and yellow components be after the solvent front had moved 8.0 cm from the origin?
- When the red component had travelled 6.0 cm from the origin, how far would the orange component have travelled?
- Sketch the chromatogram of the ink to scale after the solvent front had moved 15 cm from the origin.

- 9 The diagram below shows a thin-layer chromatogram of amino acids in a medicine.



- Which amino acids are in the medicine?
 - Amino acids are colourless. What technique could be used to make the amino acid spots visible?
 - Calculate the R_f value of taurine.
 - Which amino acid is bonded least strongly to the stationary phase?
- 10 The components in a plant extract were separated by simple column chromatography. The mobile phase used was water and the column was packed with alumina powder. Use the following diagram and select from the following terms to complete the sentences.

A	smaller	used coarser alumina powder
B	low	increased the column length
C	high	decreased the column length
	larger	



The component most strongly adsorbed to the stationary phase is band _____.

As band B begins to emerge from the column, it appears to separate into two bands. These two components would be more successfully separated if you _____.

HPLC differs from column chromatography because the particles in the stationary phase are _____ and _____ pressure is applied to the mobile phase.

Advanced applications of chromatography

- Why is it necessary to measure the peak areas produced by a number of standard solutions when performing quantitative analysis using an HPLC instrument?
 - HPLC does not directly produce measures of concentration.
 - A graph of the actual number of standard solutions against the peak area can then be drawn.
 - The peak areas are measured to ensure they remain constant throughout the analysis.
 - The less standard solutions used, the more accurate the graph will be.
- Which one of the following is the basis for the operation of HPLC?
 - Interaction of chemicals with both mobile and stationary phases
 - Movement of chemicals through a column using a carrier gas
 - Passage of chemicals through a heated column under pressure
 - Small molecules becoming trapped in gaps between solid particles in a packed column

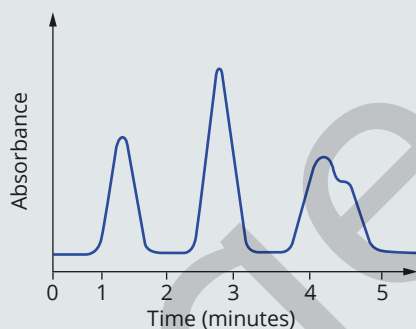
- 13 On the basis of their chemical properties, which of the following gases would be least suitable as the carrier gas for gas chromatography?

A oxygen
B argon
C nitrogen
D helium

- 14 A chemist working for an environmental agency wishes to analyse the levels of a herbicide in a country stream using HPLC. Sort the following steps in the analytical process into the correct order.

A Measure the herbicide peak areas from the chromatograms of the standard solutions and the stream water sample.
B Obtain chromatograms of the stream water and standard solutions of the herbicide.
C Construct a calibration curve and mark the herbicide peak area from the stream water sample on it.
D Use R_f values to identify the herbicide peak on the chromatogram of stream water.
E Determine the herbicide concentration in the stream water sample.

- 15 A herbal tea extract was analysed by HPLC. The chromatogram obtained is shown in the graph below.



- a Explain what information chemists can obtain from this chromatogram.
b How many components are evident?
c Briefly explain how the components are separated by the HPLC technique.

- 16 Australian wines are routinely tested for ethanol content. A quick and reliable method is by gas chromatography (GC). The peak areas produced from GC analysis of a sample of wine and a number of standard solutions of ethanol are shown in the following table.

	% ethanol	Relative ethanol peak area
Wine sample	?	5.6
Standard 1	2.0	1.6
Standard 2	4.0	3.2
Standard 3	6.0	4.8
Standard 4	8.0	6.4

- a Plot a calibration curve of relative peak area against concentration of ethanol (%).
b Determine the percentage of ethanol in the wine sample, correct to 1 decimal place.

- 17 The concentration of benzene in a sample of petrol was determined by gas chromatography. A series of standards with accurately known concentrations of benzene were run under the same conditions as the sample. Data from the chromatograms of the sample and the standards are shown in the following table. Calculate the concentration of benzene in the sample.

% benzene	Peak area (cm ²)
0.2	1.0
0.4	2.4
0.8	4.6
1.2	7.0
Sample	3.6

Connecting the main ideas

- 18 A chemist working for a local council wishes to analyse the composition of a river where children swim. It is thought the water might contain high levels of the toxic organic chemical dioxin.

Which one of the following statements concerning the analysis of this water is correct?

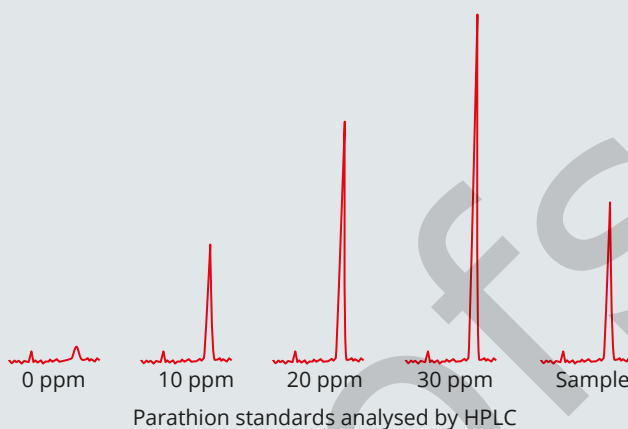
- A Paper chromatography would allow determination of the concentration of dioxin present but not confirm its identity.
B The results obtained from HPLC would provide identification and accurate concentrations of the dioxin present.
C GC can identify the dioxin present but not its concentration.
D Paper chromatography would be suitable for identifying the dioxin and measuring its concentration.

- 19** Each chromatography technique has disadvantages associated with it. Complete the table below using the options provided to indicate the problem or problems associated with each technique. Options may be used more than once.

- A** Requires relatively large amounts of solvents to operate
- B** Difficult or impossible to obtain quantitative data
- C** Expensive equipment needed
- D** Samples must be able to be dissolved in solvent

Technique	Problem
Paper or thin-layer chromatography	
High-performance chromatography (HPLC)	
Gas chromatography (GC)	

- 20** The organophosphorus insecticide parathion has been widely used in mosquito-prone areas. An empty drum of the insecticide was found close to a major reservoir. The EPA was asked to analyse the water to determine whether it was a threat to human health. Levels above 0.01 ppm (parts per million) in water are a threat to human health. Parathion has an LD_{50} value (lethal dose to 50% of test animals exposed to this concentration per kilogram of their body mass) of 8 mg kg^{-1} . Parathion standards in water of 0 ppm, 10 ppm, 20 ppm and 30 ppm parathion were prepared and analysed by HPLC, as shown in the following graph. An undiluted sample of the reservoir water was run on the column under the same conditions.



- a** Complete the table below.

Standard	Peak height (cm)
Standard 0 ppm parathion	
Standard 10 ppm parathion	
Standard 20 ppm parathion	
Standard 30 ppm parathion	
Reservoir water	

- b** Construct a calibration curve for the analysis of parathion.
- c** Determine the concentration of parathion in the water sample. (Assume that peak height is a measure of the concentration of the insecticide.)
- d** Is the reservoir water within the legal limits for safe drinking? Support your answer with a calculation.
- e** What volume of water would a laboratory mouse, mass 150g, need to drink to reach the LD_{50} dose?